

1.4 Graphs

Mathematics B (MEI) — OCR B — A-Level (H640) — Pure Mathematics — spec refs g1–g2

What this topic covers. Sketching standard curves and identifying their key features — intercepts, turning points, asymptotes — and the four graph transformations, including the vector notation MEI uses for translations.

1. What a sketch must show

A sketch is not a plot. It need not be to scale, but it must be *correct* in every labelled feature. Examiners look for:

- **Intercepts** — set $x = 0$ for the y -intercept, $y = 0$ for the x -intercepts.
- **Asymptotes** — drawn as dashed lines and labelled with their equations.
- **Turning points** — where relevant.
- **Behaviour at the extremes** — what happens as $x \rightarrow \pm\infty$.
- **Correct shape** at repeated roots.

2. Standard curves

Curve	Shape and key features
$y = x^2$	Parabola, minimum at the origin, symmetric about the y -axis.
$y = x^3$	Increasing everywhere, point of inflection at the origin, symmetric about the origin.
$y = x^4$	Like a parabola but flatter near the origin; minimum at the origin.

$y = \frac{1}{x}$	Hyperbola in two opposite quadrants; asymptotes $x = 0$ and $y = 0$.
$y = \frac{1}{x^2}$	Always positive; asymptotes $x = 0$ and $y = 0$.
$y = \sqrt{x}$	Starts at the origin, defined only for $x \geq 0$, increasing and flattening.
$y = e^x$	Always positive, through $(0, 1)$, asymptote $y = 0$ to the left.
$y = \ln x$	Defined for $x > 0$, through $(1, 0)$, asymptote $x = 0$.
$y = x $	V-shape with vertex at the origin.

3. Polynomial curves and repeated roots

Factorised form tells you almost everything. A polynomial of degree n has at most n real roots and at most $n - 1$ turning points.

Factor	Behaviour at that root
$(x - a)$	Crosses the axis straight through.
$(x - a)^2$	Touches the axis and turns back — a minimum or maximum sitting on the axis.
$(x - a)^3$	Crosses, but flattens at the crossing (point of inflection).

Worked example. Sketch $y = x^2(x - 2)$.

Roots at $x = 0$ (repeated, so the curve touches) and $x = 2$ (single, so it crosses). It is a cubic with positive leading coefficient, so $y \rightarrow -\infty$ as $x \rightarrow -\infty$ and $y \rightarrow +\infty$ as $x \rightarrow +\infty$. The curve comes up from the bottom left, touches the axis at the origin, dips, then crosses at $x = 2$.

Worked example. Find where $y = (x + 1)(x - 2)(x - 4)$ crosses the axes.

x -intercepts at $x = -1, 2, 4$. For the y -intercept set $x = 0$: $y = (1)(-2)(-4) = 8$, so $(0, 8)$.

4. Asymptotes

An **asymptote** is a line the curve approaches arbitrarily closely without meeting (at least in the limit).

- **Vertical:** where the denominator is zero and the numerator is not. For $y = \frac{1}{x-3}$, the asymptote is $x = 3$.
- **Horizontal:** found from the behaviour as $x \rightarrow \pm\infty$. For $y = \frac{x}{x-1}$, dividing top and bottom by x gives $\frac{1}{1-1/x} \rightarrow 1$, so $y = 1$.
- **Oblique (slant):** arises when the numerator's degree is one more than the denominator's. For $y = \frac{x^2+1}{x} = x + \frac{1}{x}$, the term $\frac{1}{x} \rightarrow 0$, so the curve approaches $y = x$.

Worked example. State the asymptotes of $y = \frac{2}{x+1} - 3$.

Vertical: $x = -1$. As $x \rightarrow \pm\infty$, the fraction tends to 0, so the horizontal asymptote is $y = -3$.

A curve *may* cross a horizontal or oblique asymptote — the asymptote only describes behaviour far out. It can never cross a vertical asymptote, because the function is undefined there.

Watch for a cancelling factor. In $y = \frac{x^2-1}{x-1} = x+1$ there is no asymptote at $x = 1$ — instead there is a **hole** in the line, because the function is undefined at that single point.

5. Transformations

Four transformations, and the crucial split between those acting on the **output** (outside the function) and those acting on the **input** (inside).

Transformation	Effect on $y = f(x)$	Acts on
----------------	----------------------	---------

$y = f(x) + a$	Translation a units up	Output
$y = f(x + a)$	Translation a units left	Input
$y = af(x)$	Stretch, factor a , parallel to the y -axis	Output
$y = f(ax)$	Stretch, factor $\frac{1}{a}$, parallel to the x -axis	Input
$y = -f(x)$	Reflection in the x -axis	Output
$y = f(-x)$	Reflection in the y -axis	Input

Everything **inside** the bracket behaves in the way you do not expect: $f(x + 2)$ moves the graph *left*, and $f(2x)$ *halves* the width. Everything outside behaves as expected. This single rule resolves most transformation errors.

Vector notation for translations

MEI writes a translation as a column vector. The graph $y = f(x - 2) + 5$ is the graph of $y = f(x)$ translated by

$$\begin{pmatrix} 2 \\ 5 \end{pmatrix}$$

that is, 2 units in the positive x -direction and 5 in the positive y -direction. Note the sign: $f(x - 2)$ shifts *right* by 2.

Tracking a point

Worked example. $y = f(x)$ has a maximum at $(2, 5)$. Find the maximum of $y = 3f(x + 1) - 2$. Work through the transformations in order:

- $f(x + 1)$: translate 1 left, so x becomes $2 - 1 = 1$.
- $3f(\cdot)$: stretch factor 3 vertically, so y becomes $5 \times 3 = 15$.
- -2 : translate 2 down, so y becomes 13.

The maximum is at $(1, 13)$.

Order of combined transformations

Worked example. Describe $y = 2f(3x - 6)$ as a sequence of transformations.

First rewrite the inside in factorised form: $3x - 6 = 3(x - 2)$, so $y = 2f(3(x - 2))$.

- Translate 2 units in the positive x -direction.
- Stretch parallel to the x -axis, factor $\frac{1}{3}$.
- Stretch parallel to the y -axis, factor 2.

Always factorise the inside first. Reading $f(3x - 6)$ as "translate 6 then stretch $\frac{1}{3}$ " gives the wrong graph, because the stretch would also scale the translation. Writing $3(x - 2)$ makes the correct order unambiguous.

6. Intersections and roots

The number of real solutions of $f(x) = g(x)$ is the number of intersections of $y = f(x)$ and $y = g(x)$. Sketching both curves often answers "how many solutions" far faster than algebra.

Worked example. How many real solutions does $x^3 - 3x = k$ have when $-2 < k < 2$?

The curve $y = x^3 - 3x$ has turning points where $3x^2 - 3 = 0$, i.e. $x = \pm 1$, giving a local maximum at $(-1, 2)$ and a local minimum at $(1, -2)$. A horizontal line $y = k$ strictly between -2 and 2 cuts the curve three times.

Worked example. How many times do $y = e^x$ and $y = x + 1$ meet?

Exactly once, at $x = 0$. The line is the tangent to the exponential curve at that point, and $e^x \geq x + 1$ everywhere.

7. Symmetry

Condition	Symmetry	Example
$f(-x) = f(x)$	Symmetric about the y -axis (even)	$y = x^2, y = \cos x$
$f(-x) = -f(x)$	Symmetric about the origin (odd)	$y = x^3, y = \sin x$

8. Common pitfalls

- Translating the wrong way for $f(x + a)$ — it moves **left**, not right.
- Using stretch factor a instead of $\frac{1}{a}$ for $f(ax)$.
- Failing to factorise the inside before describing a combined transformation, so the order comes out wrong.
- Drawing a curve crossing a *vertical* asymptote.
- Missing that a repeated factor makes the curve touch rather than cross the axis.
- Missing a cancelling factor and inventing an asymptote where there is only a hole.
- Leaving asymptotes unlabelled, or omitting intercepts, on a sketch that asks for key features.